

EV CYBER ACADEMY

LINUX & LAB COMMAND REFERENCE

Mission-by-Mission Commands • Outputs • Definitions

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Environment: Kali Linux + EV Lab

Target: 192.168.56.103

This document records the commands used during the authorized four-mission EV Cyber Academy laboratory, with their purpose and observed results.

Mission 1 — FTP Anonymous Access

Command

```
nmap -p 21 192.168.56.103
```

Definition: Scans TCP port 21 for the FTP service.

Output / Result: 21/tcp open ftp; vsFTPD 3.0.3 identified.

Command

```
ftp 192.168.56.103
```

Definition: Starts the FTP client and connects to the target.

Output / Result: 220 (vsFTPD 3.0.3).

Command

```
anonymous
```

Definition: FTP username used to test anonymous/guest access.

Output / Result: Anonymous login succeeded in the completed lab.

Command

```
ls
```

Definition: Lists files in the remote FTP directory.

Output / Result: The lab evidence/flag file was listed.

Command

```
get flag.txt
```

Definition: Downloads flag.txt from the FTP server.

Output / Result: Local copy created.

Command

```
cat flag.txt
```

Definition: Displays the downloaded file.

Output / Result: FLAG{FTP_ANONYMOUS_ACCESS_LEAKED_SUCCESSFULLY}.

Mission 2 — Web Command Injection / Defensive Validation

Command

```
ip -4 addr
```

Definition: Shows IPv4 addresses on target interfaces.

Output / Result: The lab target used 192.168.56.103/24.

Command

```
systemctl status apache2 --no-pager
```

Definition: Checks Apache service status without an interactive pager.

Output / Result: Apache was active/running.

Command

```
cp /var/www/html/ping.php /var/www/html/ping.php.bak
```

Definition: Creates a backup before editing the PHP file.

Output / Result: ping.php.bak was created.

Command

```
cat /var/www/html/ping.php
```

Definition: Displays PHP source for inspection.

Output / Result: Hardened version used FILTER_VALIDATE_IP and escapeshellarg().

Command

```
curl -i http://192.168.56.103/...
```

Definition: Sends an HTTP request and shows headers/body.

Output / Result: Invalid-IP testing showed the lab's Defensive Alert message.

Command

```
curl -I http://192.168.56.103/
```

Definition: Requests HTTP headers only for quick verification.

Output / Result: HTTP response/status was returned.

Mission 3 — Apache Directory Indexing

Command

```
grep -n "Options" /etc/apache2/sites-available/0000-default-lab.conf
```

Definition: Searches the lab virtual-host configuration for Options directives.

Output / Result: Configuration showed Options -Indexes +FollowSymLinks after the fix.

Command

```
grep -Rni "Options +Indexes" /etc/apache2 2>/dev/null
```

Definition: Recursively searches Apache configuration for indexing directives; stderr is redirected.

Output / Result: Used to locate the relevant configuration.

Command

```
apache2ctl configtest
```

Definition: Validates Apache configuration syntax.

Output / Result: Syntax OK.

Command

```
systemctl restart apache2
```

Definition: Restarts Apache to apply the configuration.

Output / Result: Apache restarted successfully.

Command

```
find /var/www/html -maxdepth 2 -type d
```

Definition: Lists web directories up to two levels deep.

Output / Result: backups, admin and secret_configs were found.

Command

```
curl -i http://192.168.56.103/backups/
```

Definition: Tests the backups directory.

Output / Result: 403 Forbidden; no directory listing.

Command

```
curl -i http://192.168.56.103/secret_configs/
```

Definition: Tests the secret_configs directory.

Output / Result: 403 Forbidden; no directory listing.

Command

```
curl -i http://192.168.56.103/admin/
```

Definition: Tests the admin directory.

Output / Result: 200 OK because an index/page resource was served.

Mission 4 — SMB Anonymous Share

Command

```
sudo apt update
```

Definition: Refreshes Kali's package index.

Output / Result: Package index updated.

Command

```
sudo apt install smbclient -y
```

Definition: Installs the Samba SMB client; -y accepts the prompt.

Output / Result: smbclient became available.

Command

```
smbclient --version
```

Definition: Shows the installed SMB client version.

Output / Result: Confirmed installation.

Command

```
smbclient -L //192.168.56.103 -N
```

Definition: Lists SMB shares without a password prompt.

Output / Result: EV_Academy_Share was listed.

Command

```
smbclient //192.168.56.103/EV_Academy_Share -N
```

Definition: Connects to the authorized share anonymously.

Output / Result: smb: \> prompt appeared.

Command

```
ls
```

Definition: Lists files inside the SMB share.

Output / Result: confidential_flag.txt was found.

Command

```
get confidential_flag.txt
```

Definition: Downloads the evidence file.

Output / Result: Local file downloaded.

Command

```
exit
```

Definition: Leaves the smbclient shell.

Output / Result: Returned to Kali prompt.

Command

```
cat confidential_flag.txt
```

Definition: Displays the downloaded file.

Output / Result: FLAG{SMB_ANONYMOUS_ACCESS_COMPROMISED_2026}.

Quick Reference

Tool / Command	Purpose	Mission
nmap	Port/service discovery	1
ftp	FTP connection and transfer	1
ip / ip link	Network interface inspection	1–2
systemctl	Service status/restart	2–3
grep	Search configuration/text	2–3
cp	Create backups	2
cat	Read files	1–4
curl	HTTP verification	2–3
find	Find/list directories	3
apache2ctl configtest	Check Apache syntax	3
smbclient	SMB enumeration/access	4
apt update/install	Package management	4

Common Terminal Issue: dquote>

When Bash shows dquote>, it is waiting for a missing closing double quote. Press Ctrl+C to cancel the unfinished command and return to the normal shell prompt. For example, enter curl -I http://192.168.56.103/ without unnecessary quotation marks.

Defensive Lessons

The lab demonstrated the risks of anonymous FTP/SMB access, unsafe web input reaching shell execution, and Apache directory indexing. Recommended defenses include disabling anonymous access, strict input validation and safe argument handling, and using Options -Indexes where directory contents should not be exposed.

Completion Record

Mission 1: Completed • Mission 2: Completed • Mission 3: Completed • Mission 4: Completed • Overall: 4/4 missions completed.